probability, and hence 1/λ is the average life. Instead of the latter, the *half-life* period, *T* (that period of time during which exactly one-half of the original number of atoms decays) is frequently used. The half-life is slightly less than the average life; it differs from the latter by the factor log nat 2, the natural logarithm of 2. (If we write $t = \frac{1}{\lambda} \log \text{ nat } 2$, then $N = N_0 \times e^{-\log \text{ nat } 2} = \frac{1}{2} N_0$).This law applies to both alpha and beta radiation.

Thus the radioactive properties of a homogeneous substance are determined principally by two factors: the nature of the emitted particles and the average life or half-life of the substance.

The gamma rays play a somewhat different part. We must point out, first of all, that in natural radioactivity gamma rays do not appear alone, but only in combination with one of the other two types of radiation. Gamma rays have an even greater penetrating power than either the x-rays (to which they are essentially analogous in other respects) or the alpha and beta rays. To give an approximate idea of their power of penetration, let it be said that while an alpha ray is absorbed by a single sheet of paper, 100 such sheets would be required to absorb a beta ray, and several thick volumes to absorb the gamma radiation. As already mentioned, gamma rays cannot be deflected, nor—unlike the alpha and beta rays—can they be made visible in the cloud chamber. For although the gamma rays too cause ionization in air, this is not a direct, primary process, but an indirect ionization through the agency of the electrons which the gamma rays dislodge. In the cloud chamber, we can see only the tracks of the secondary particles generated by them; the tracks of the gamma rays themselves are not visible. These two facts are quite compatible with each other, since both can be attributed to the absence of an electric charge in gamma rays.

Actually, gamma rays differ from x-rays, and even from visible light, too, merely in that the wavelength of the gamma rays is much shorter. They are a species of electro-magnetic waves, among which the radio waves have the longest wavelength. At any rate, we know that the already mentioned wave-particle duality applies to all these types of radiation. Hence, while we have just referred to gamma rays as an electromagnetic wave issuing forth from the atomic nucleus, we may just as well look at them under their particle aspect and speak of particles, extremely energetic photons, ejected from the nucleus and speeding through space with the velocity of light.

The fact that under certain conditions atomic nuclei can emit gamma rays, is quite understandable. We already know that the extranuclear structure of the atom can emit light when—for example—the atom is excited due to a gaseous discharge. Atoms emit x-rays also when particles are dislodged from their inner electron shells by extremely fast electrons. This follows clearly from the fact that the extranuclear structure of the atom constitutes an